

LAB 2

Agile Backlog Creation & Sprint Simulation in Jira

OpenAPI Mock Server Generator

Name	Sriviswaja M
SRN	PES1UG24AM288
Jira Space	AM288
Space Key	SCRUM
Date	1 September 2026

Submitted as evidence of Jira backlog setup, story estimation, sprint simulation, burndown analysis, and reflection.

1. Objective and Required Deliverables

The objective of this lab was to convert the requirements from Lab 1 into Agile Epics and User Stories, prioritize and estimate the backlog, simulate Scrum sprints in Jira, and analyse progress using burndown charts.

Required evidence: Jira backlog with Epics and User Stories; story point assignments; sprint board/Active Sprint view; burndown chart; and a brief reflection answering the four specified questions.

2. Agile Backlog Structure

The Jira project represents an OpenAPI Mock Server Generator. The backlog was organized into three major functional Epics, with User Stories assigned to the appropriate Epic.

Epic	User Stories	Story Points
Open API Specification Processing	SCRUM-9 — Import OpenAPI 3.0 Specification SCRUM-10 — Parse and Validate OpenAPI Specification	$3 + 5 = 8$
Mock Endpoint & Response Generation	SCRUM-11 — Generate Dynamic Mock Endpoint SCRUM-12 — Generate Schema-Compliant JSON Payloads	$8 + 8 = 16$
API Testing & Simulation	SCRUM-13 — Configure Simulated Response Latency SCRUM-14 — Validate API Requests and Responses	$3 + 5 = 8$

The six User Stories visible in the final backlog were assigned Fibonacci-style story points of 3, 5, and 8. The Sprint 1 workload shown in Jira totals 16 story points ($3 + 5 + 8$), while the remaining three stories in the backlog total another 16 points ($8 + 5 + 3$).

3. Jira Evidence Screenshots

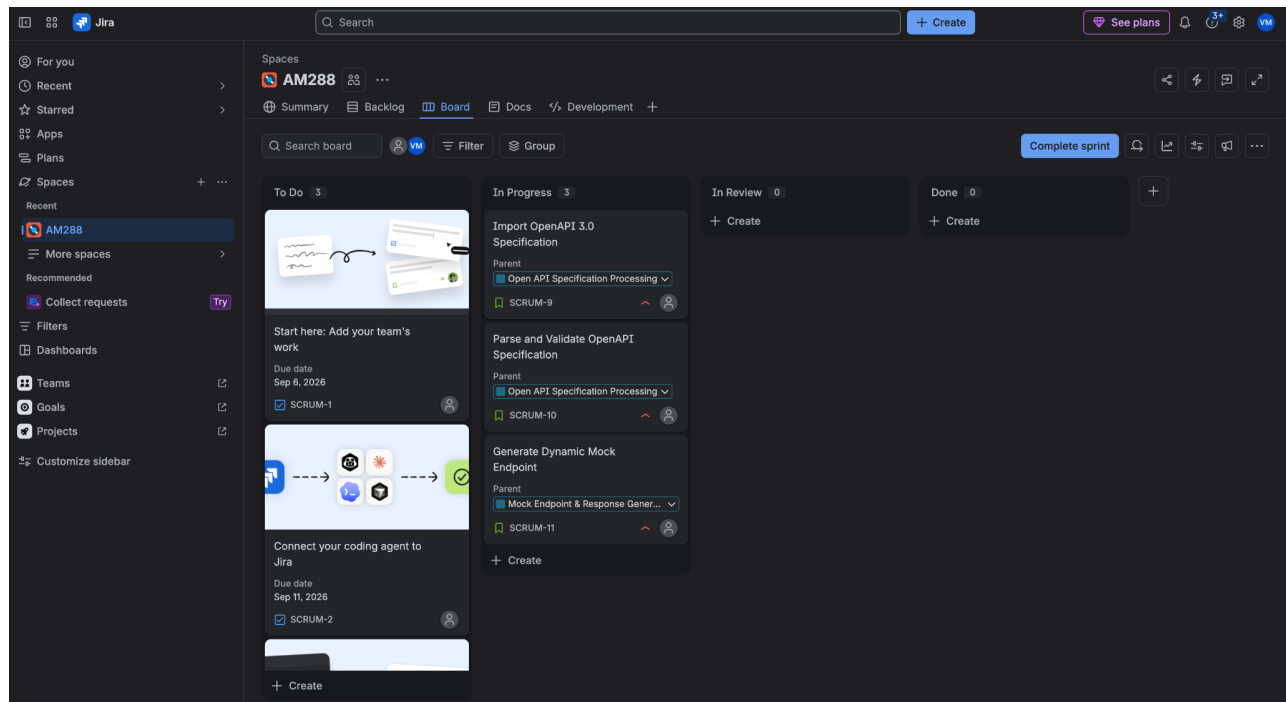


Figure 1. Jira board/backlog setup and initial sprint state.

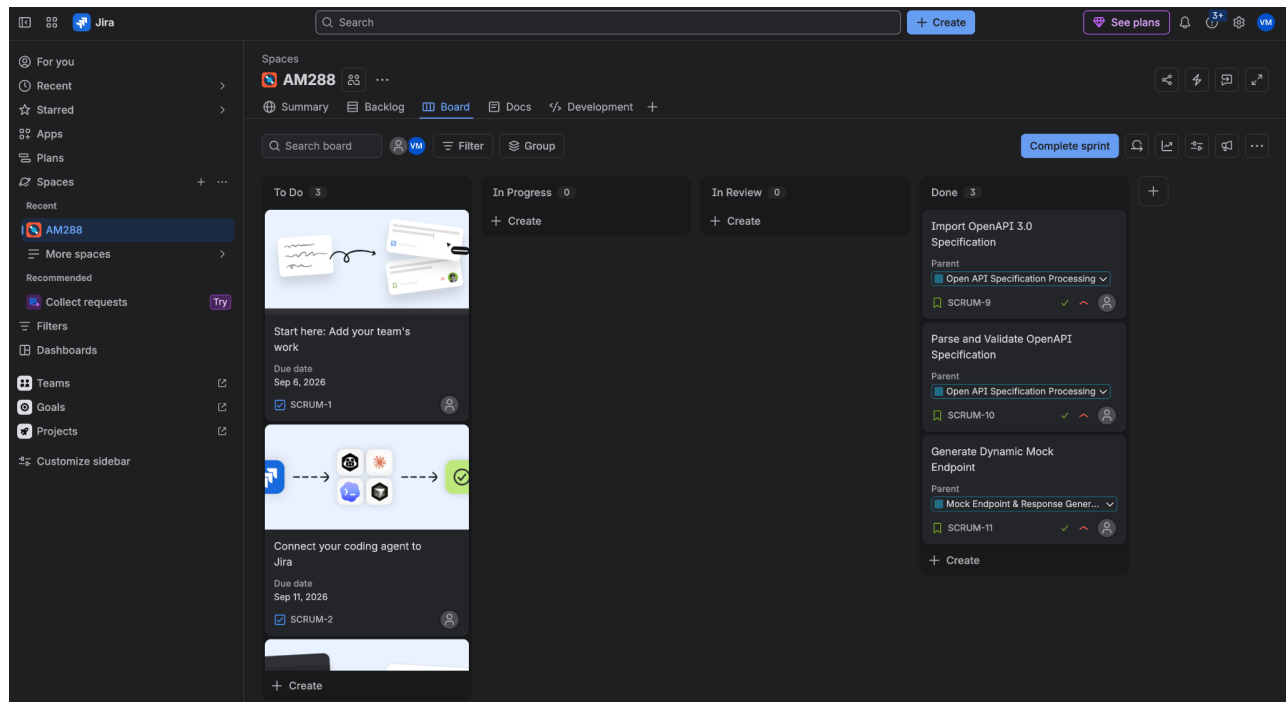


Figure 2. Jira board showing completed Sprint 1 work.

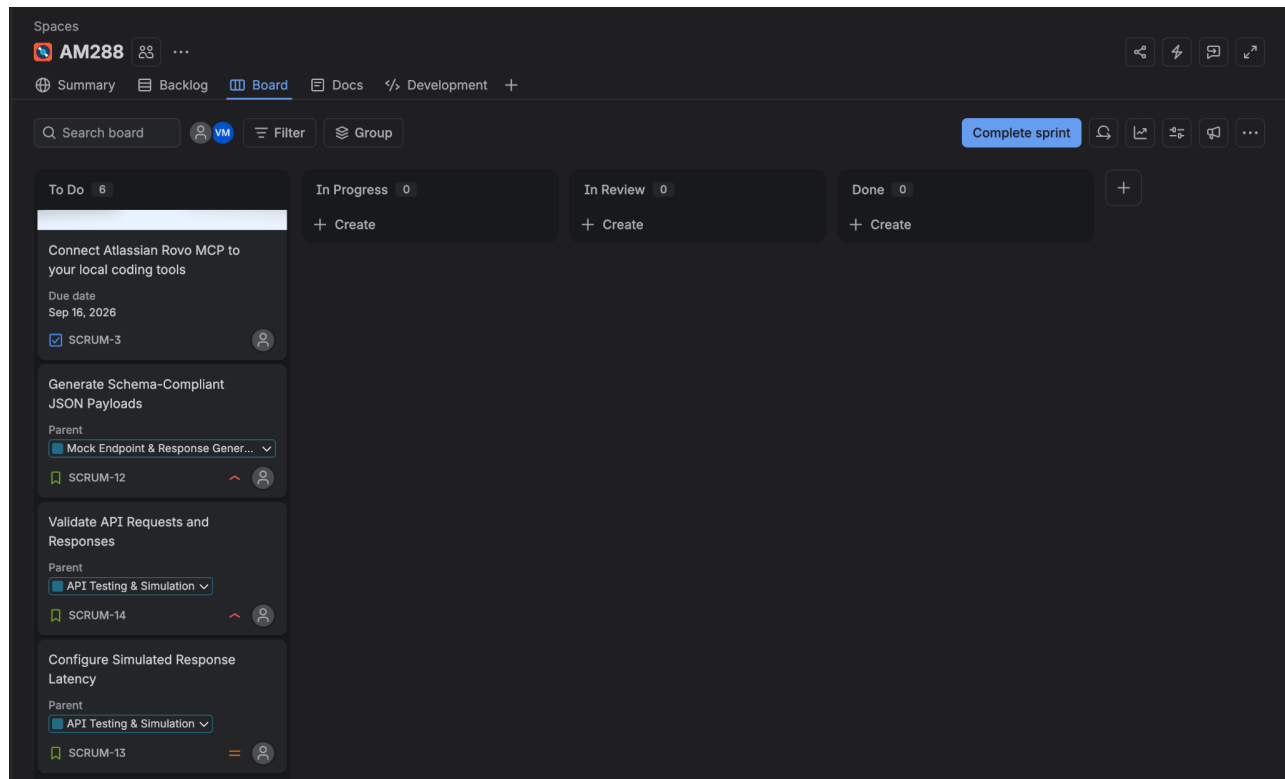


Figure 3. Sprint 2 board with remaining stories being progressed.

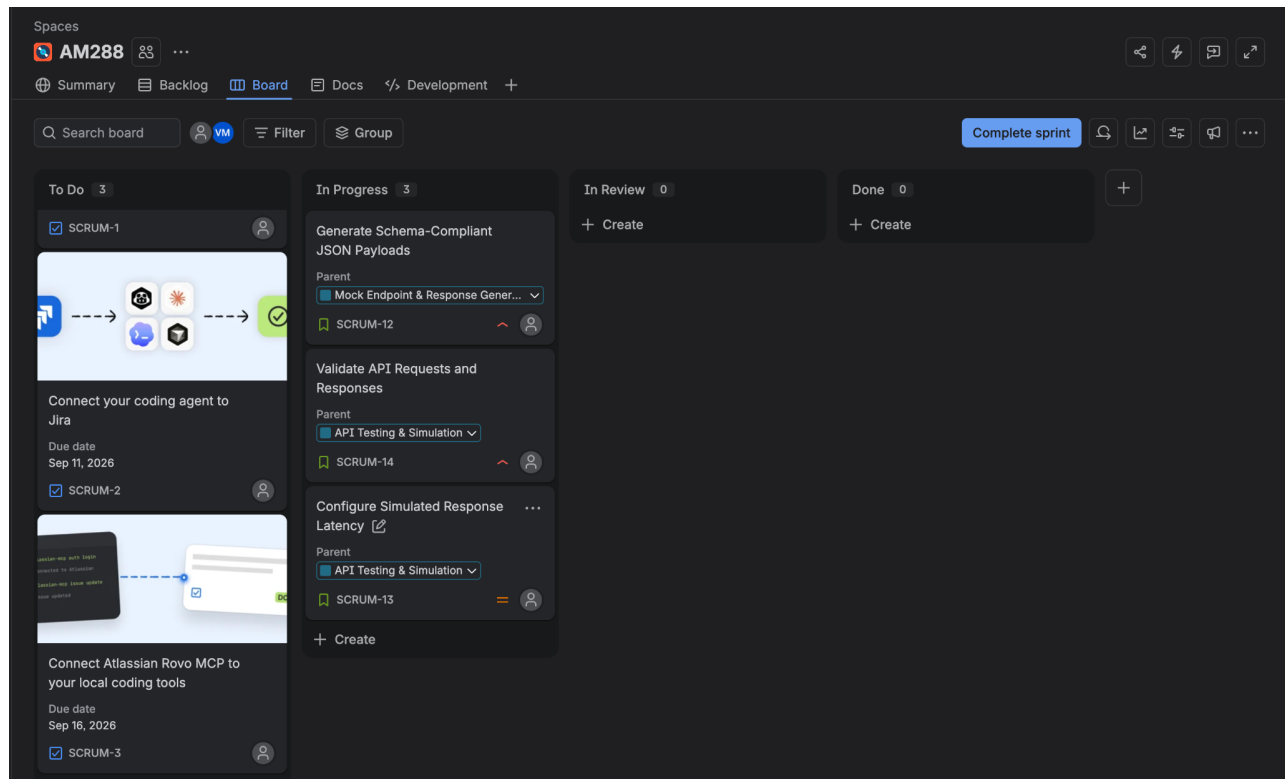


Figure 4. Sprint 2 board after the three selected stories were moved to Done.

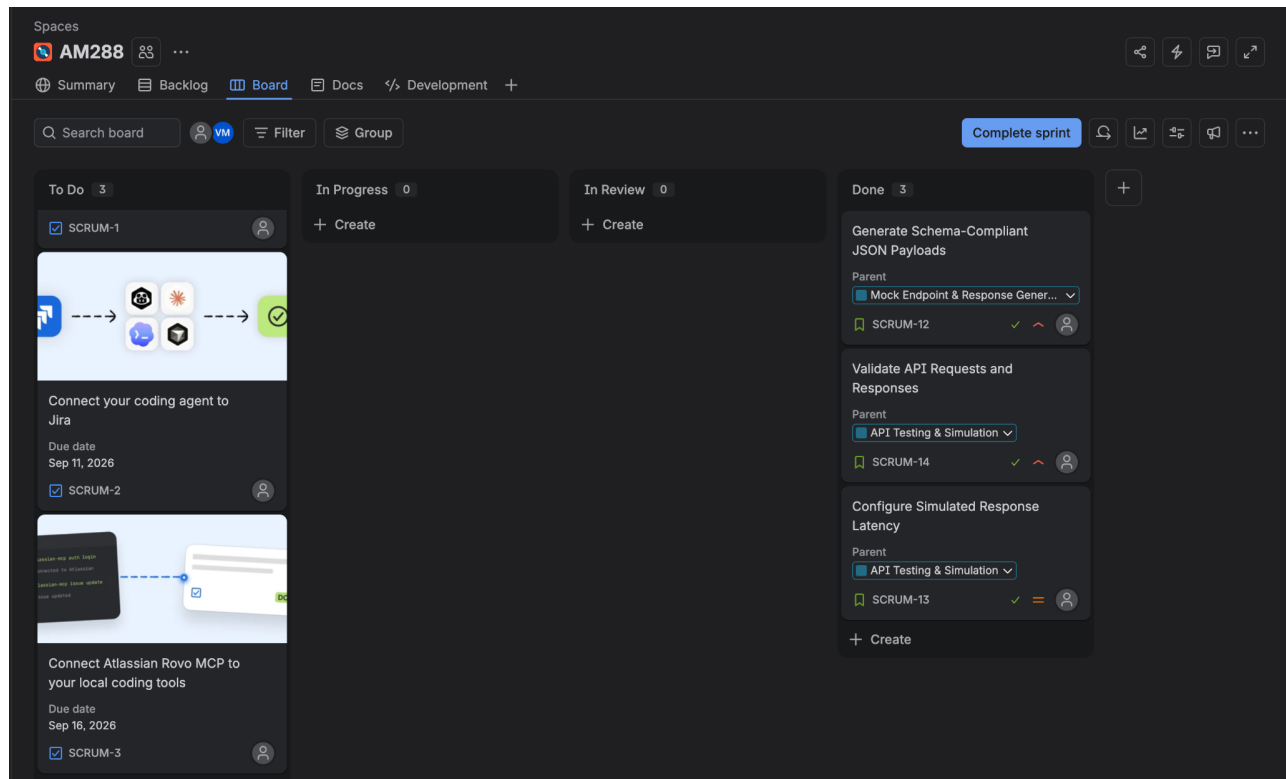


Figure 5. Sprint 2 burndown chart using Story Points.

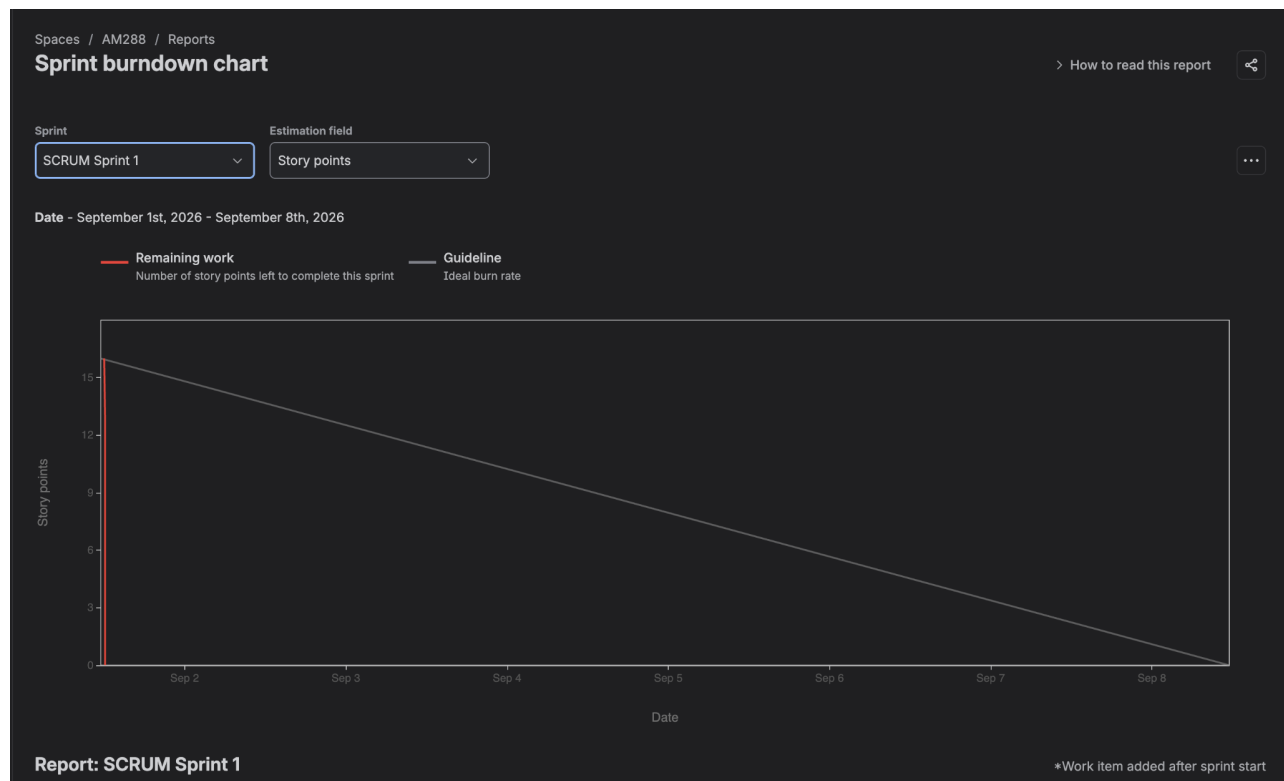


Figure 6. Sprint 1 burndown chart using Story Points.

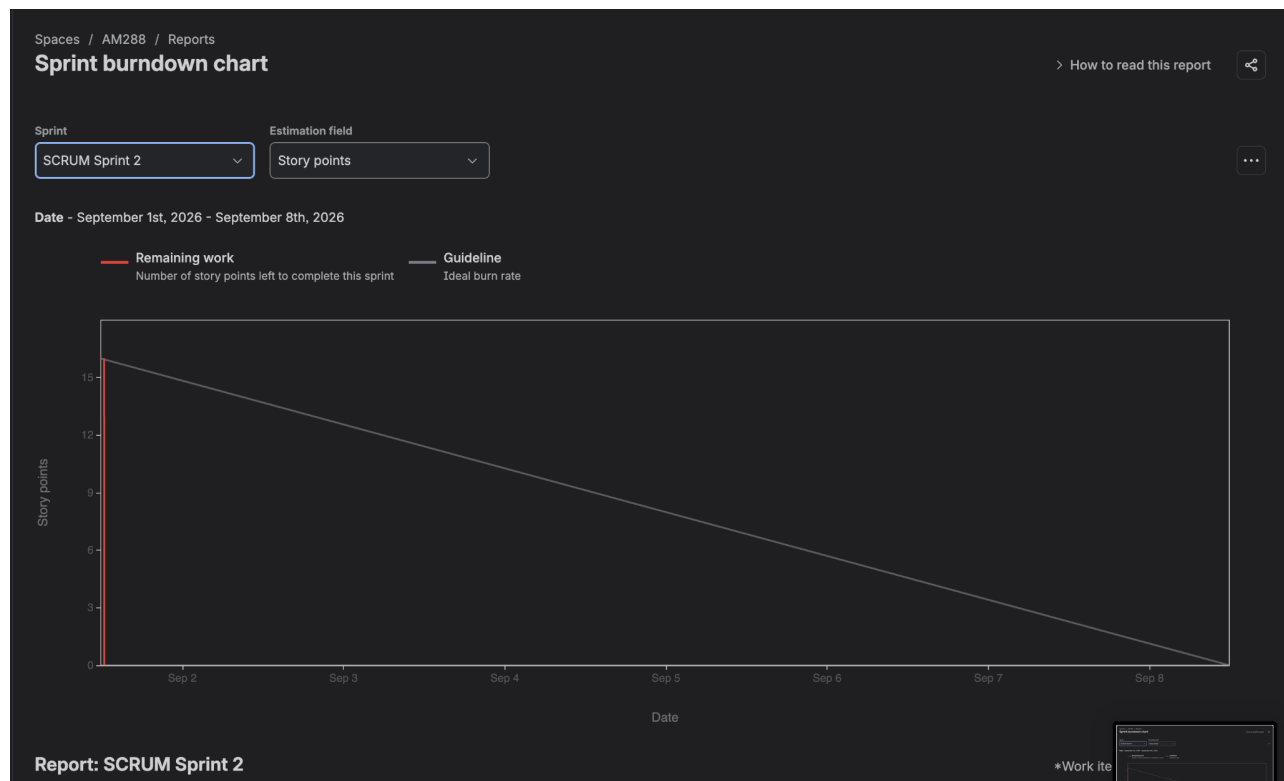


Figure 7. Sprint 2 burndown chart using story points.

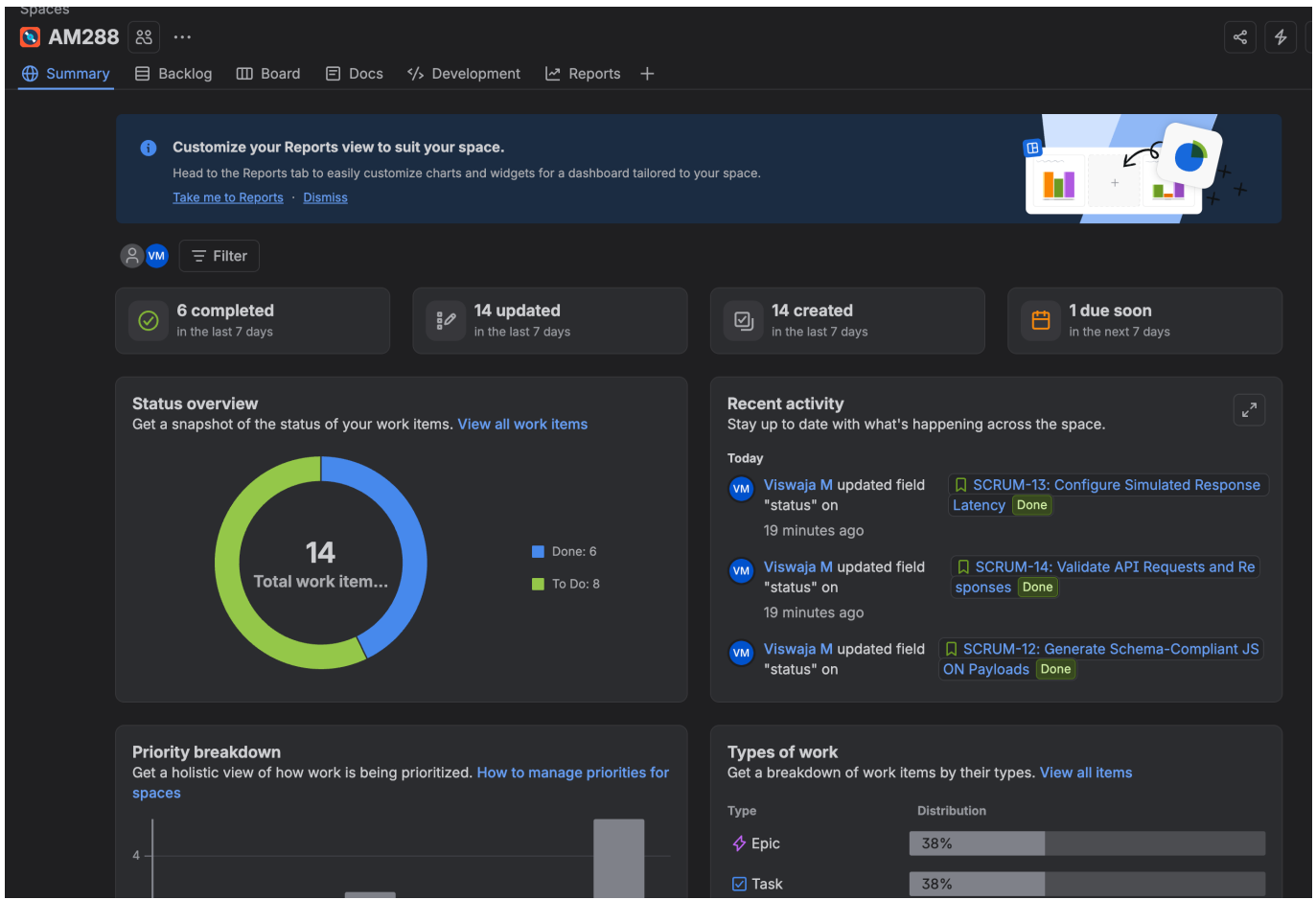


Figure 8. Final summary

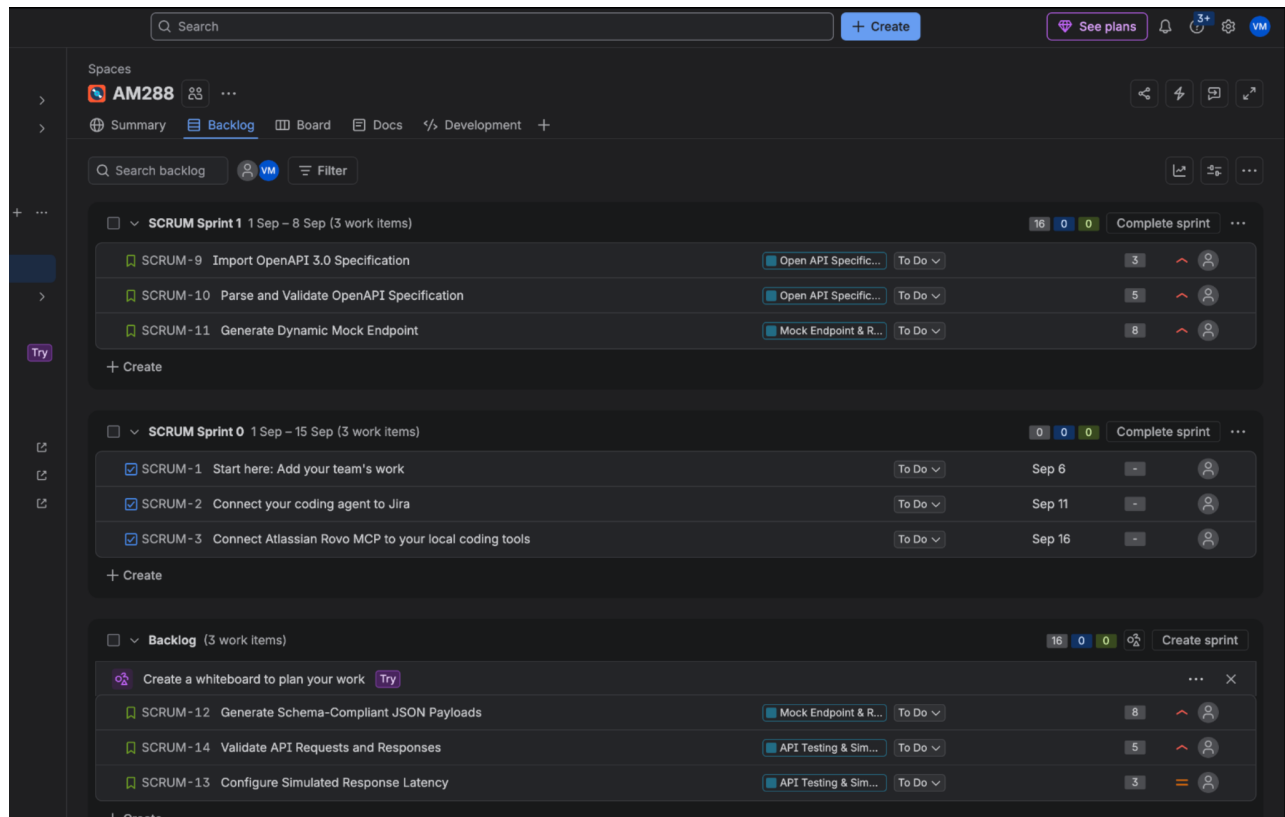


Figure 8. Additional Jira evidence screenshot supplied with the submission.

4. Sprint Execution Summary

Two one-week sprint simulations were configured. The Jira evidence shows the first sprint containing SCRUM-9, SCRUM-10, and SCRUM-11, with a total of 16 story points. The second sprint used the remaining stories SCRUM-12, SCRUM-14, and SCRUM-13, also totaling 16 story points.

Sprint	Stories	Points	Initial State	Final State
Sprint 1	SCRUM-9, SCRUM-10, SCRUM-11	16	To Do	Done
Sprint 2	SCRUM-12, SCRUM-14, SCRUM-13	16	To Do	Done

The board evidence demonstrates the Scrum workflow by moving User Stories through To Do, In Progress, and Done. Both sprint sets were completed in the simulation.

5. Reflection

1. Did your estimations reflect the actual effort?

Yes. The story-point estimates provided a reasonable representation of the relative effort, complexity, and uncertainty of the tasks. Larger stories such as generating mock endpoints and schema-compliant payloads were assigned 8 points, while smaller tasks were assigned 3 or 5 points. The Fibonacci-style scale helped differentiate the relative size of the work.

2. Was your backlog well-prioritized?

Yes. The backlog was organized around the main functional areas of the OpenAPI Mock Server Generator. Core functionality such as OpenAPI processing, mock endpoint generation, payload generation, and API validation was prioritized before the response-latency simulation task. This provides a logical order based on the functional importance of the system.

3. How did your simulated sprint align with your plan?

The simulated sprints aligned well with the planned workflow. Sprint 1 focused on OpenAPI specification processing and dynamic mock endpoint generation, while Sprint 2 addressed schema-compliant payload generation, API validation, and response-latency simulation. The stories were progressed through the Jira workflow from To Do to In Progress to Done.

4. What insights did the burndown chart give about your team's capacity?

The burndown charts show that the planned 16 story points for each sprint were completed. The remaining-work line drops sharply to zero rather than following the ideal guideline gradually. This indicates that the simulated team had sufficient capacity for the selected workload and that work was completed much earlier than the ideal burn rate. In a real project, consistently finishing this early could indicate that the team is capable of committing to more work in future sprint planning; a very sharp drop can also suggest that tasks were completed or estimated in a way that did not reflect a steady distribution of effort.

6. Conclusion

The Jira workspace was configured with Epics, User Stories, priorities, story-point estimates, and two simulated one-week sprints. The sprint boards demonstrate task progression to Done, and the burndown charts provide evidence of completed story points and team capacity. The exercise demonstrates the use of Jira to apply basic Agile/Scrum backlog planning, estimation, sprint execution, and retrospective analysis.